

Dynamic Probabilistic Forecasting Update of Solar and Wind Energy Generation

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Why update a forecast in the middle of the day?

- ▶ Grid operators schedule power plants using a **day-ahead (DA) forecast**, issued the evening before and covering the whole next day.
- ▶ As the day unfolds, realized generation **differs** from the DA forecast.
- ▶ Operators are increasingly pivoting to **out-of-market** interventions (i.e., uplift payments) to maintain system reliability.

Our question

- ▶ How can be a DA forecast **revised during the day**?
- ▶ Can the update be fully **probabilistic**, not just a new point forecast?
- ▶ Can one method serve for **any asset and time**, without retraining the model?



The idea in one picture

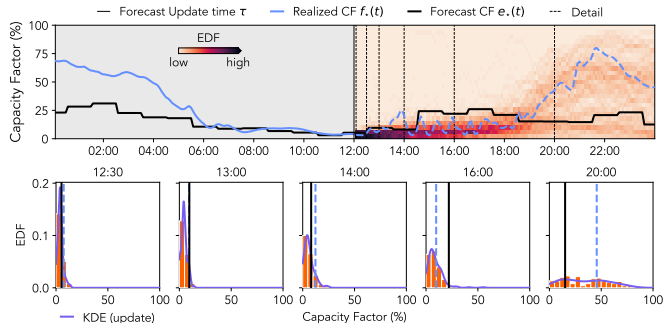


Figure 1: Generation during the morning (blue) was above the DA forecast (black).

- **Forecast update time:** **fusing** observed generation + DA forecast + similar historical **curves** → a **predictive distribution** for the rest of the day;
- with a **better** (i.e., narrower) estimation of the **uncertainty** in the generation for the evening.

What we work with: a collection of *curves*

- ▶ Every asset-day has a **generation** *curve* observed every 5-minute;
- ▶ and a hourly **DA forecast** *curve*.
- ▶ ERCOT (Texas) VRE fleet, 2017–2018:
 - ▶ **187 wind** farms
 - ▶ **211 solar** farms
- ▶ **No weather or parametric models** —the historical curves *are* the model.

Functional Data Analysis

Each trajectory is treated as a **single realization**, rather 288 discrete samples.

A **curve** inherits shape and temporal dependence (i.e., ramps and persistence) from observed trajectories.



An ensemble of plausible afternoons

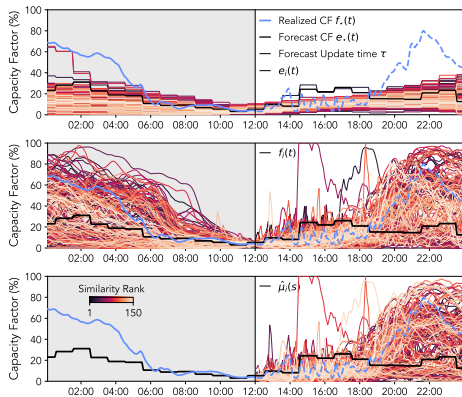
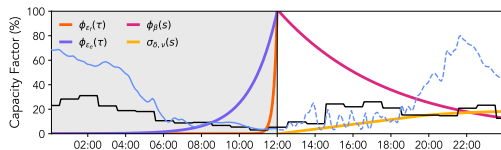
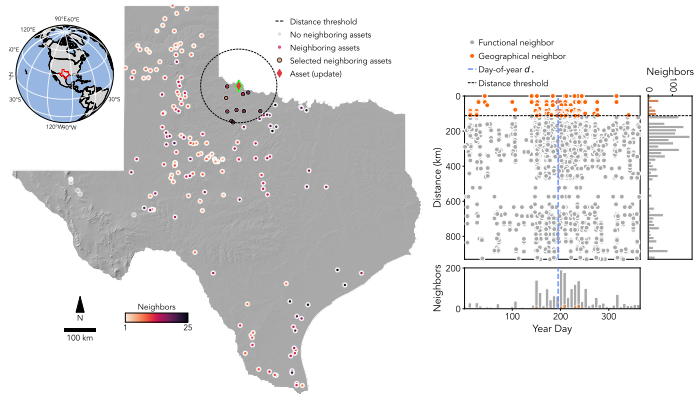


Figure 2: Functional k -nearest neighbors.



- ▶ A morning tells us **a lot** about the next few hours, but **less** about tonight.
- ▶ So each ensemble member is **blended toward the DA forecast** as we look further ahead.
- ▶ Days with 0% or 100% generation may skip blending entirely to avoid “**zero- or one-inflated**” distributions.

Find historical days that look like today



- ▶ A candidate day must be close to today in **all three** channels at once: the **partially observed curve**, the **DA forecast**, and the **day-of-year**.
- ▶ Then keep only **geographically nearby** assets (dashed circle).

Figure 3: Find historical days that look like today, only from nearby assets.

Adaptive uncertainty quantification

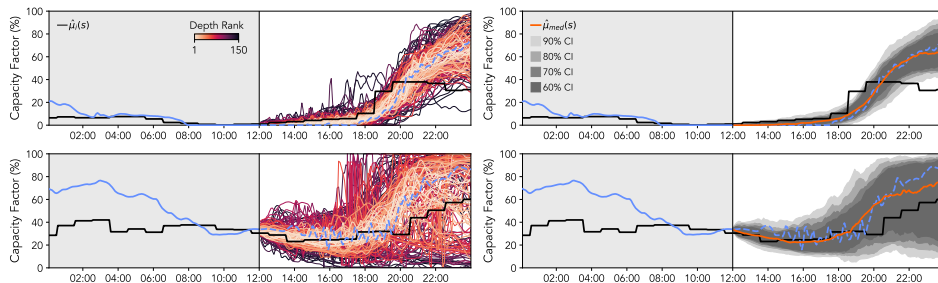


Figure 4: The selected neighbors form an ensemble of 150 real trajectories.

- ▶ **Left:** a typical day — many close matches, the ensemble agrees.
- ▶ **Right:** an unusual day — few good matches, the ensemble fans out. **Uncertainty grows exactly when it should.**

The forecast updates as the day unfolds

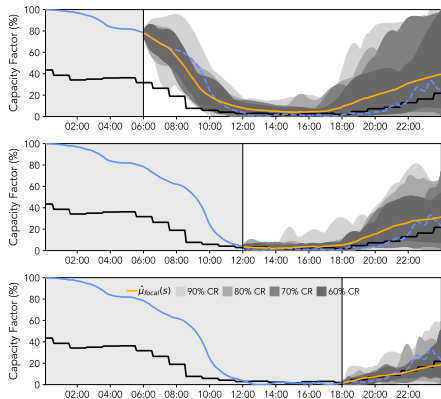


Figure 5: Focal envelope at 6am/12pm/6pm.

- ▶ Bands tighten as observations became available.
- ▶ **Bayesian optimization** minimizes **Kolmogorov–Smirnov distance** (PIT), with an **accuracy penalty** so precision is not sacrificed.
- ▶ **Calibration check**: if the forecast distributions are right → the **PIT histogram** should be flat.

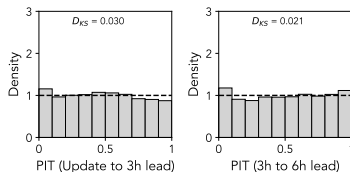


Figure 6: Out-of-sample PIT histograms: nearly flat.

The same approach works for solar generation

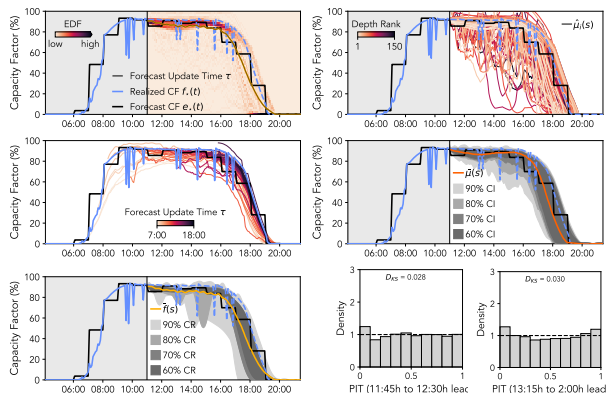


Figure 7: Solar generation forecast update at 11am.

- ▶ Two solar-specific tweaks:
 - ▶ compare days only over the **daylight window**, so long overnight zeros don't fake similarity;
 - ▶ measure season by the **sun's position** (equinox-aware), not calendar distance.
- ▶ **Widening bands** toward sunset, we achieve nearly **flat** PIT histograms.

Conclusions

Take Aways

- ▶ A **nonparametric, training-free** way to turn any day-ahead forecast into a calibrated intra-day probabilistic forecast.
- ▶ Improves **calibration without giving up accuracy**, and produces interpretable scenarios, confidence bands, and functional envelopes.
- ▶ Data-light and asset-agnostic.

Limitations

- ▶ The geographic distance is blind to **terrain, climate zones, and local weather regimes**.
- ▶ Operational costs are **asymmetric**, so perfect calibration may warrant intentional bias.



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